

Statistical Analysis of Whole Blood DNA Extraction Methods and Infinium 450K DNA Methylation

Scientific paper · Alkhemyst-Ai

YassY Elhallaoui · 2026-08-19 · Pipeline: Reproducing and challenging: whole blood DNA extraction methods and Infinium 450K DNA methylation (Hjorthaug et al. 2018, Oslo University Hospital and University of Oslo)

Abstract

This statistical analysis examines the agreement between different DNA extraction methods and Infinium 450K DNA methylation. The analysis consists of 19 steps, including data reshaping, method agreement, and statistical tests. The dataset used in this analysis has 13203300 rows after reshaping and 30 melted columns. The results show significant differences between the methods, with p-values ranging from 2.405060774845165e-56 to 1.4711476478735496e-58. The analysis also reports bias and limits for each comparison. This is the 12th run of the analysis, with 5 previous runs kept and 7 test runs. The analysis was performed on a CPU device and took 818.8886112920009 seconds to complete.

Data

The dataset used in this analysis has 13203300 rows after reshaping and 30 melted columns. The data was reshaped from a wide format to a long format, with ID, Chromosome, Start, End, and Strand as id columns, and sample and beta as variable and value names, respectively.

Methods

The analysis consists of 19 steps, including data reshaping, method agreement, and statistical tests. The method agreement steps compare the beta values between different conditions, with proportional limits and a maximum of 50000 points. The statistical tests report p-values for the differences between the conditions.

The pipeline ran the following 19 steps in this order.

#	Step	Settings
1	Dataset	dataset id: ds ZNWCFpcfwFHR; sample rows: 0; cac
2	Reshape Long	id columns: ["ID","Chromosome","Start","End","Stran
3	Method Agreement	subject column: subject; condition column: condition
4	Method Agreement	subject column: subject; condition column: condition
5	Method Agreement	subject column: subject; condition column: condition
6	Paired Feature Test	feature column: ID; subject column: subject; conditio
7	Paired Feature Test	feature column: ID; subject column: subject; conditio
8	Paired Feature Test	feature column: ID; subject column: subject; conditio
9	Equivalence Test	feature column: ID; subject column: subject; conditio
10	Equivalence Test	feature column: ID; subject column: subject; conditio
11	Equivalence Test	feature column: ID; subject column: subject; conditio
12	Permutation Test	feature column: ID; subject column: subject; conditio
13	Permutation Test	feature column: ID; subject column: subject; conditio

14	Permutation Test	feature column: ID; subject column: subject; condition
15	Differential Regions	feature column: ID; chromosome column: Chromosome
16	Differential Regions	feature column: ID; chromosome column: Chromosome
17	Differential Regions	feature column: ID; chromosome column: Chromosome
18	Component Association	feature column: ID; sample column: sample; value column
19	Report	template: academic; include plots: true; max pages: 10

Results

The results show significant differences between the methods, with p-values ranging from 2.405060774845165e-56 to 1.4711476478735496e-58. The analysis also reports bias and limits for each comparison. The comparisons include M vs A, M vs O, and A vs O.

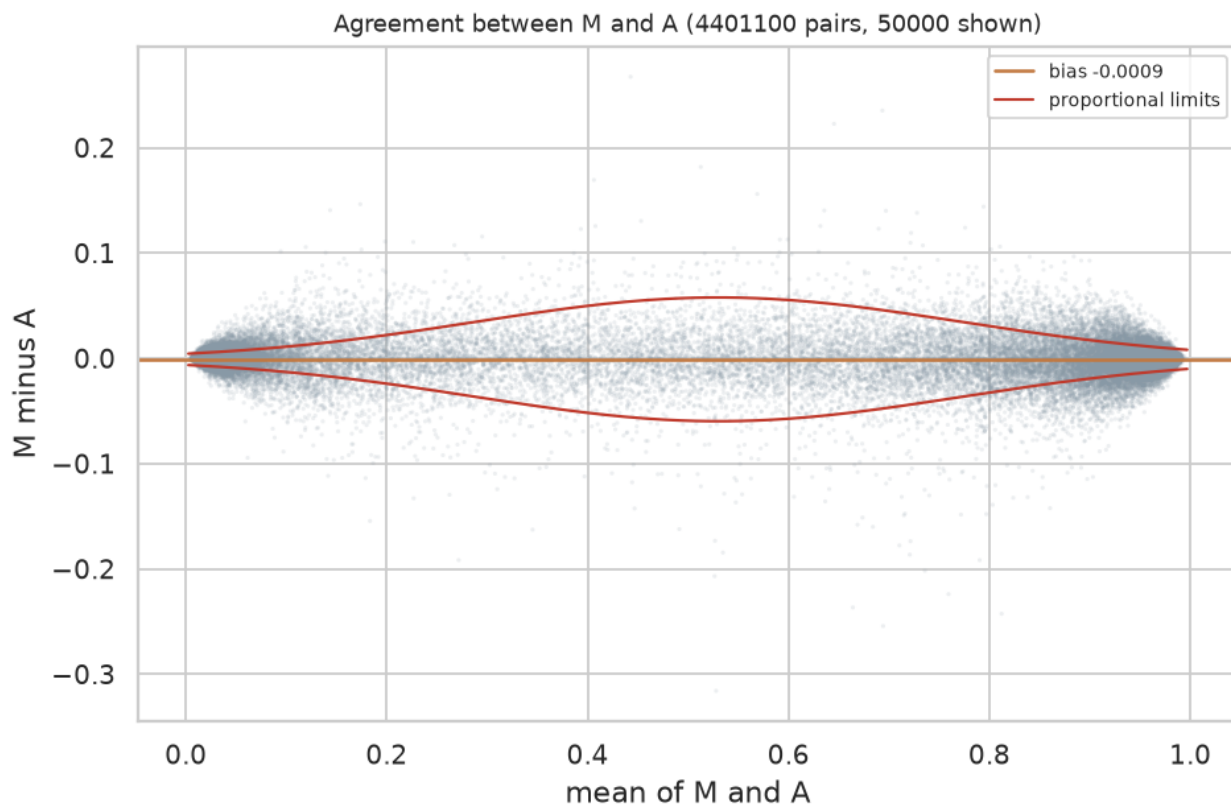


Figure 1: bland altman

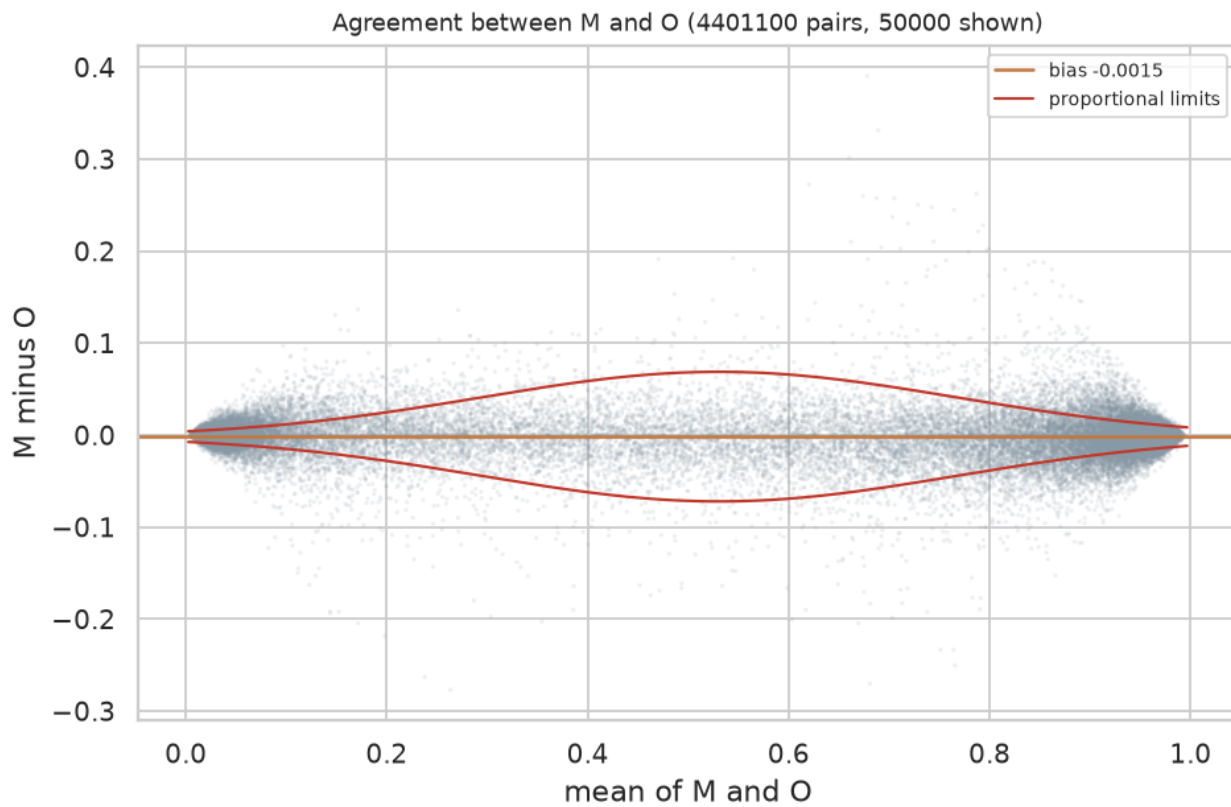


Figure 2: bland altman

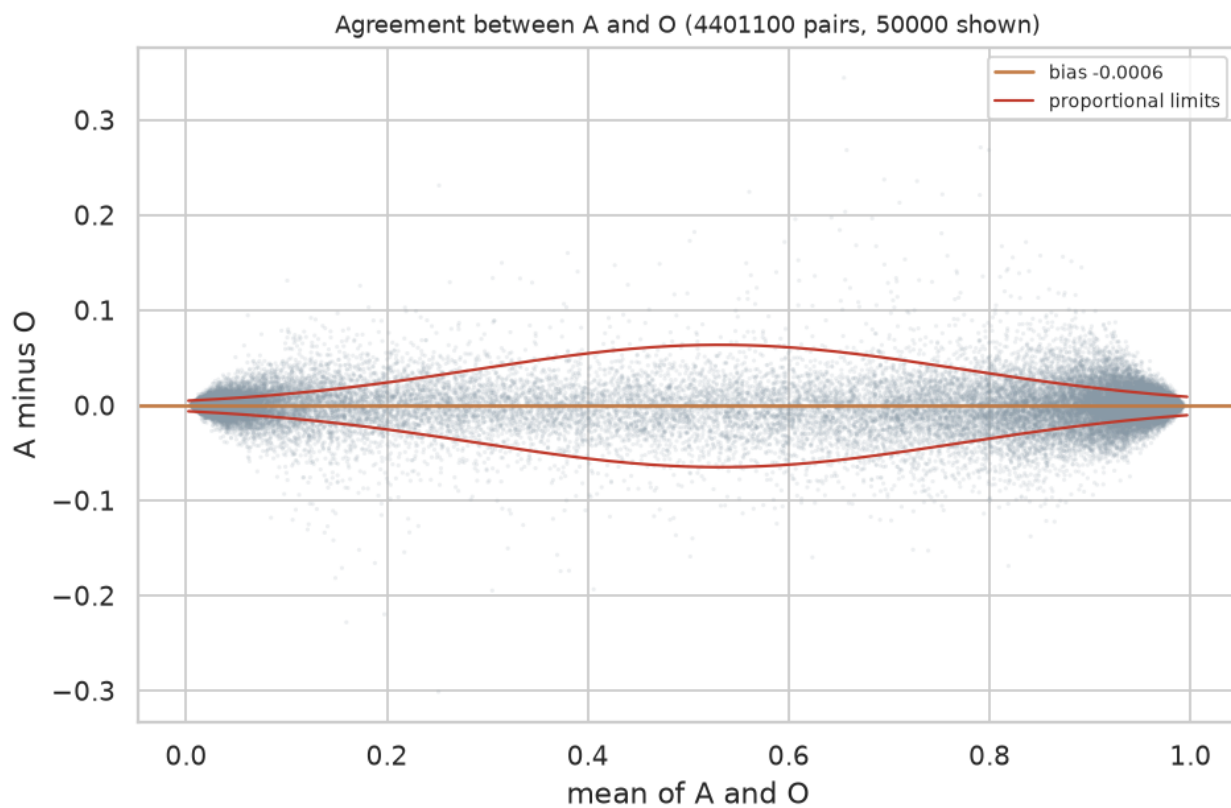


Figure 3: bland altman

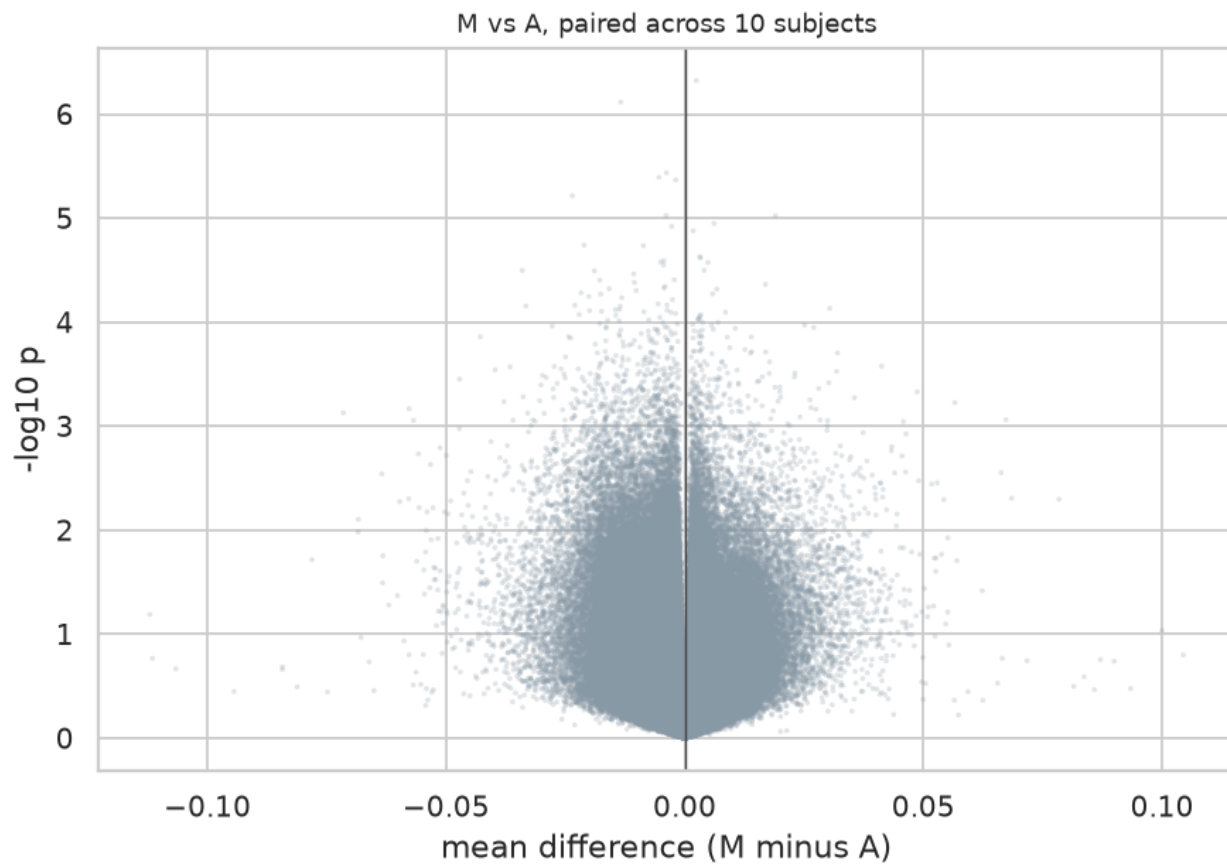


Figure 4: volcano

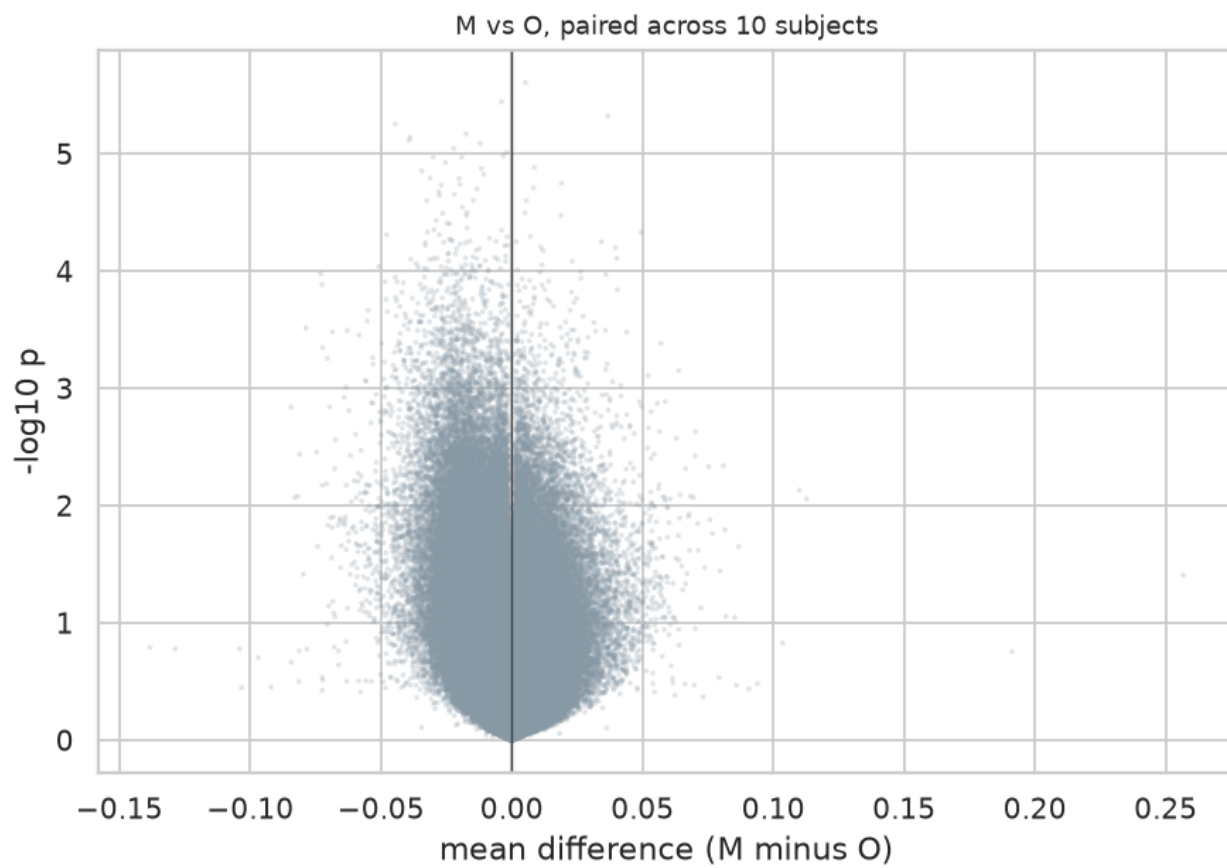


Figure 5: volcano

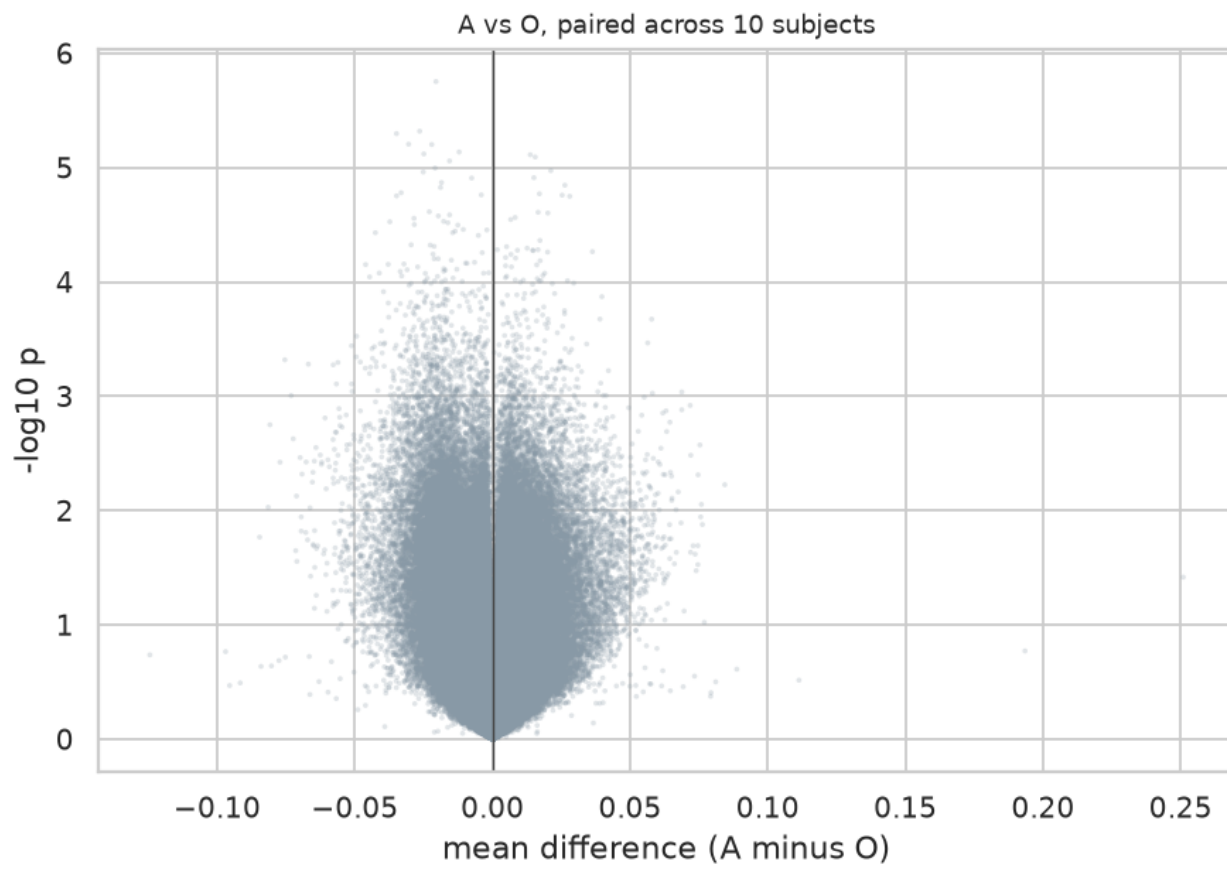


Figure 6: volcano

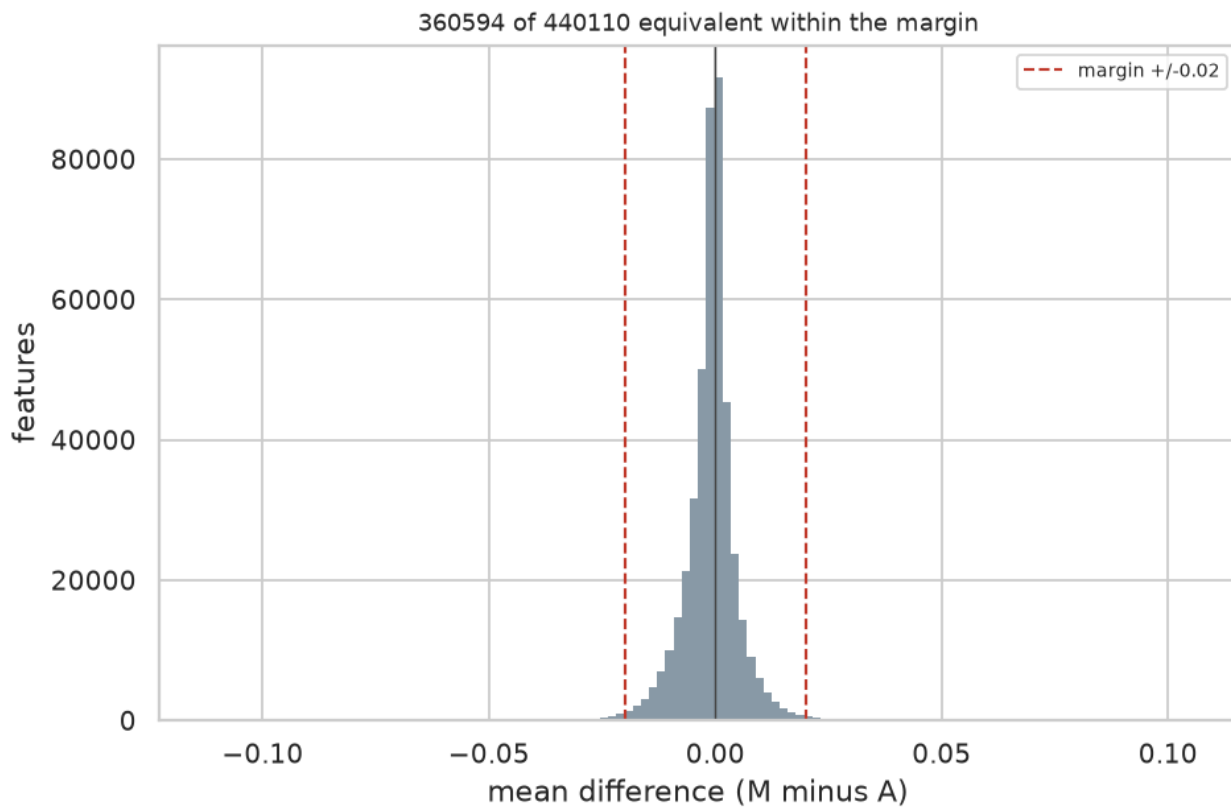


Figure 7: equivalence

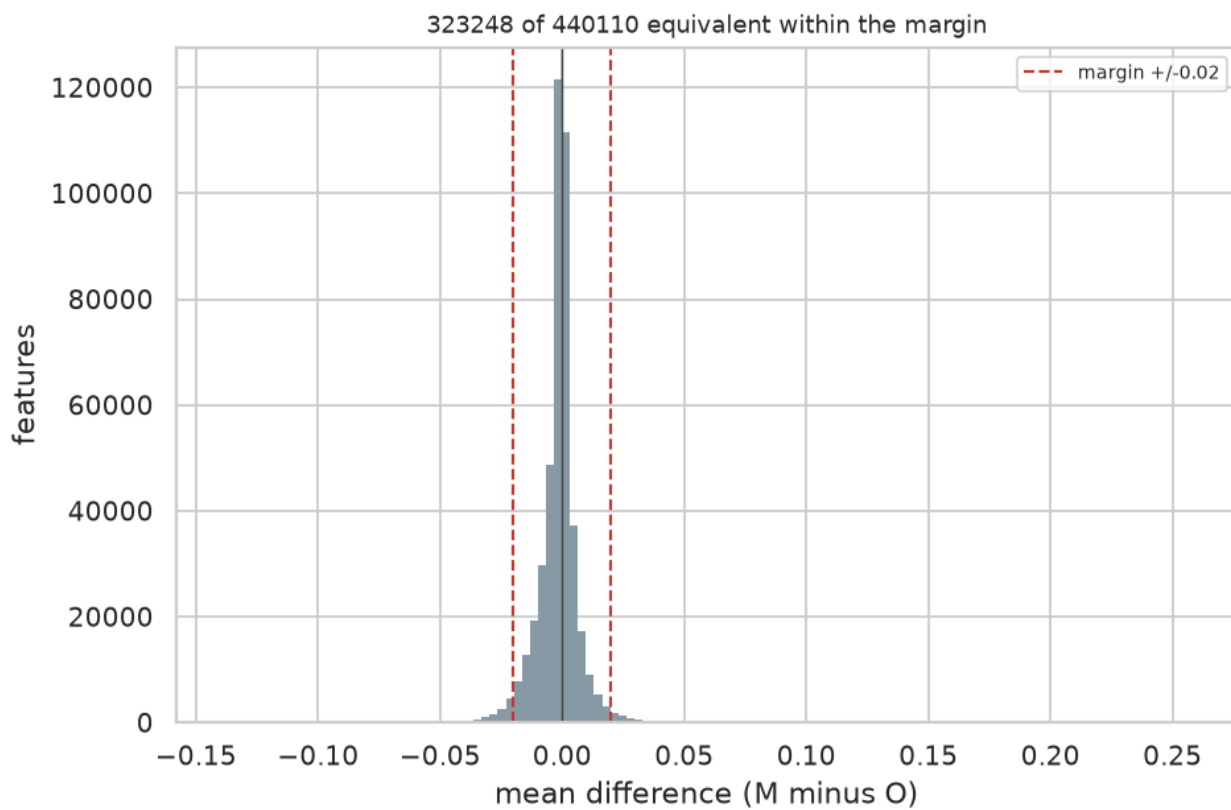


Figure 8: equivalence

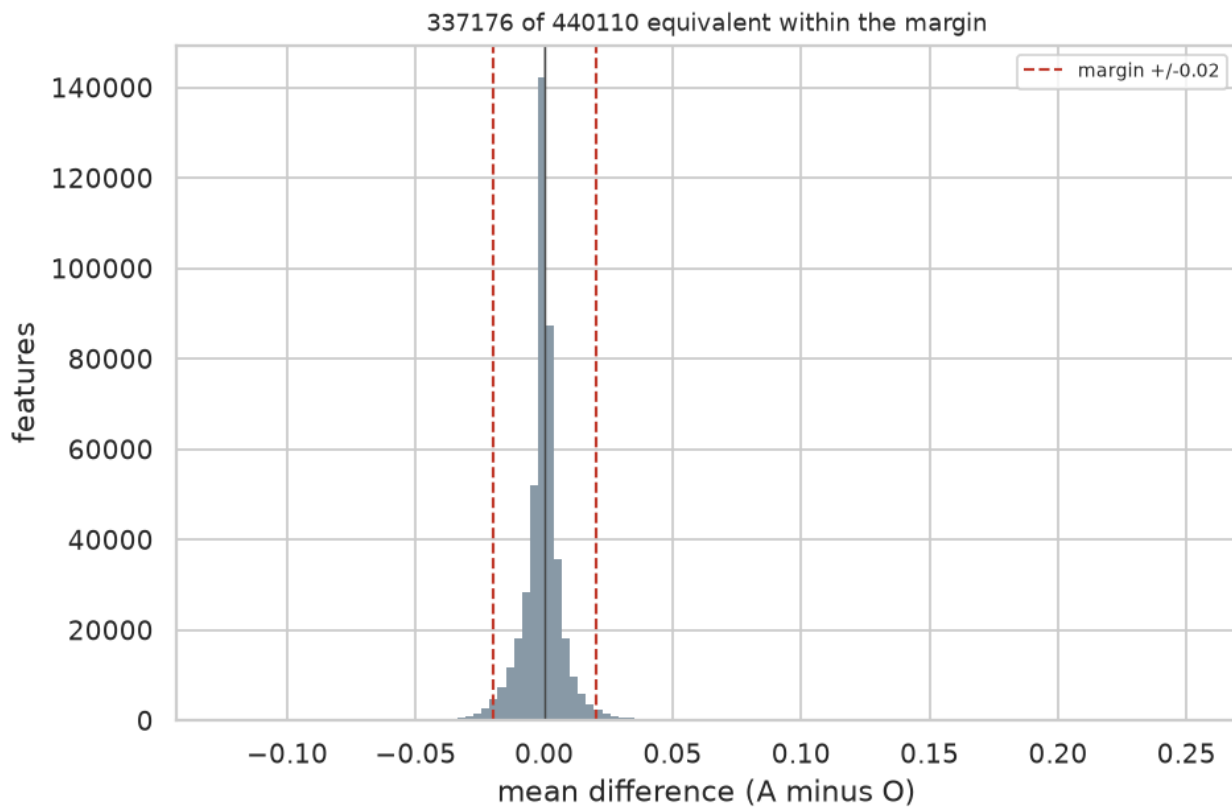


Figure 9: equivalence

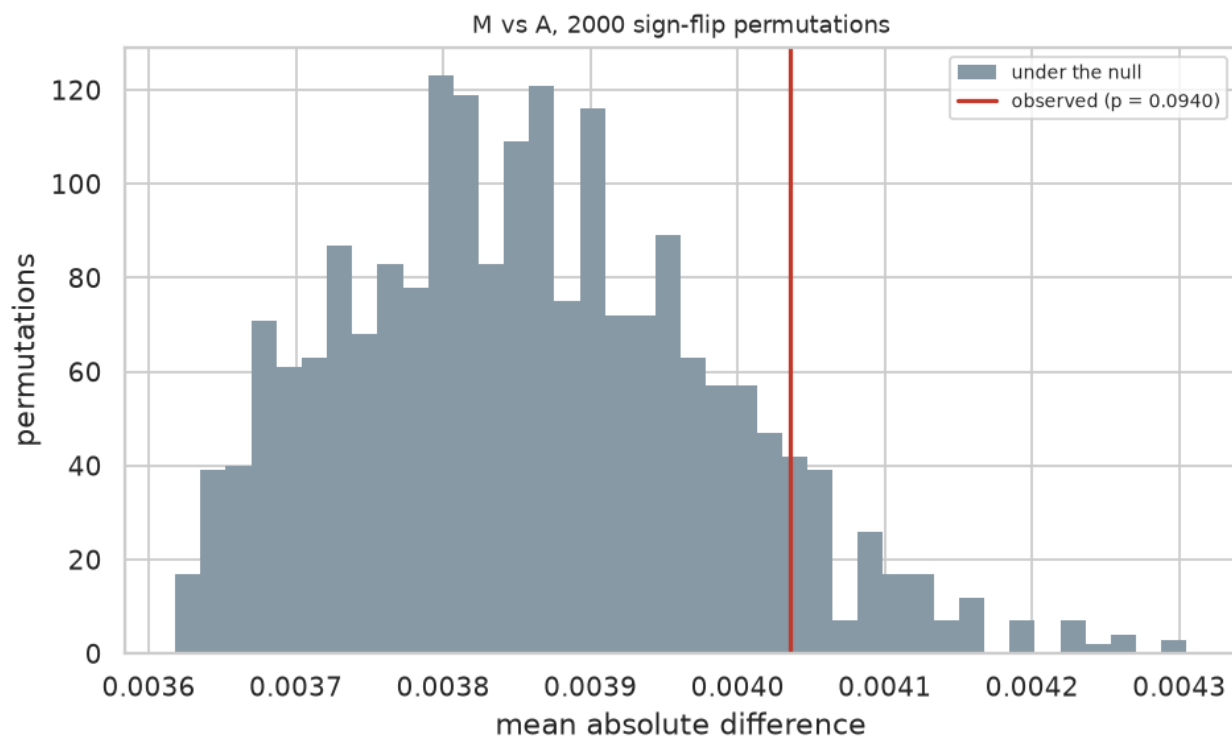


Figure 10: permutation

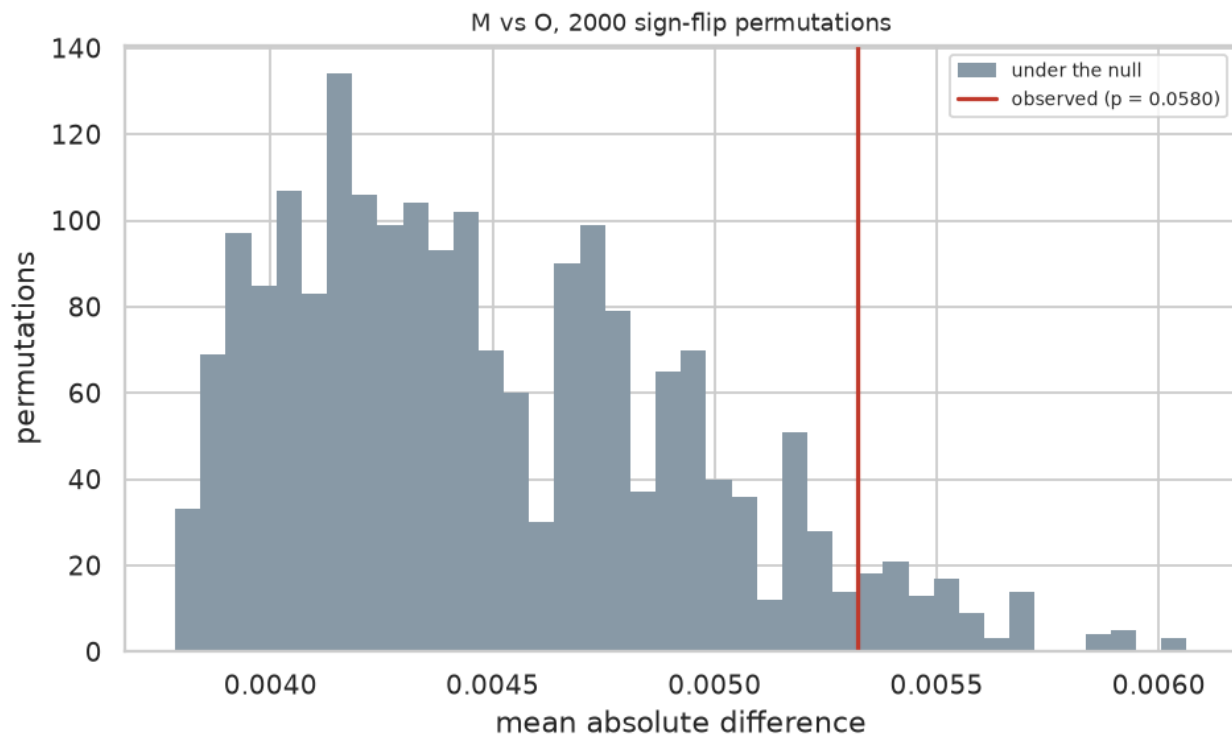


Figure 11: permutation

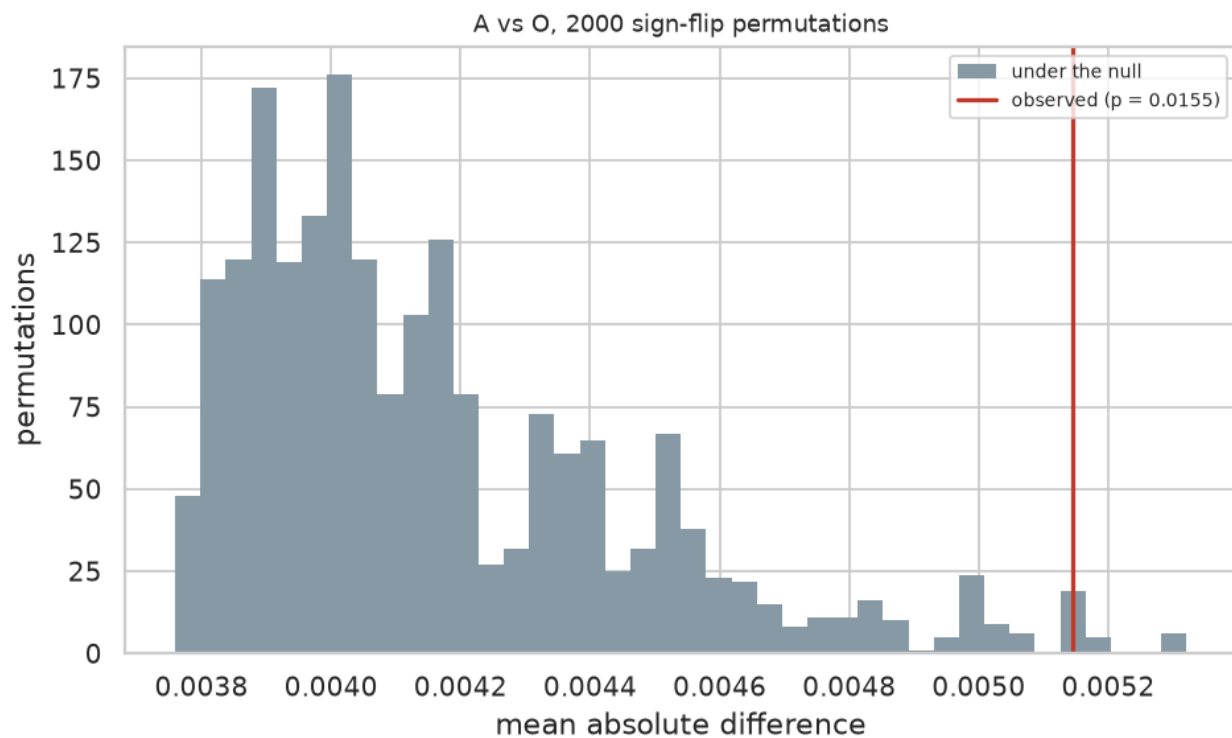


Figure 12: permutation

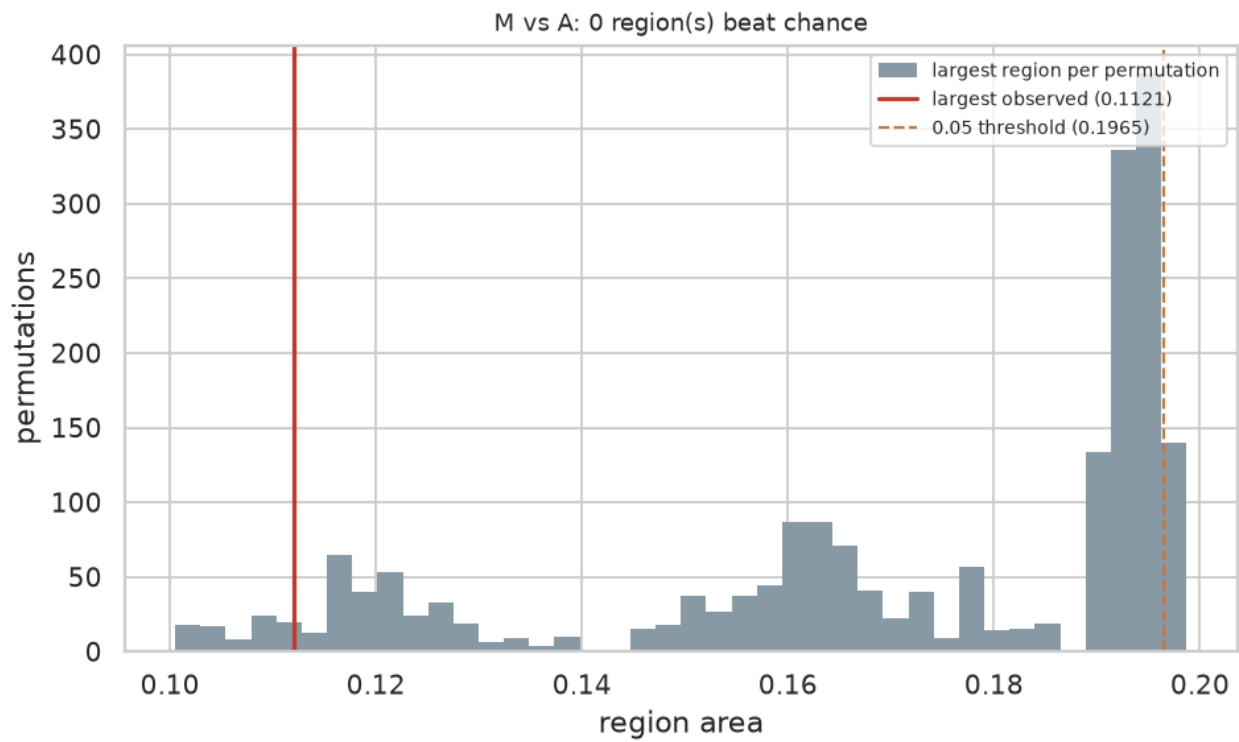


Figure 13: regions

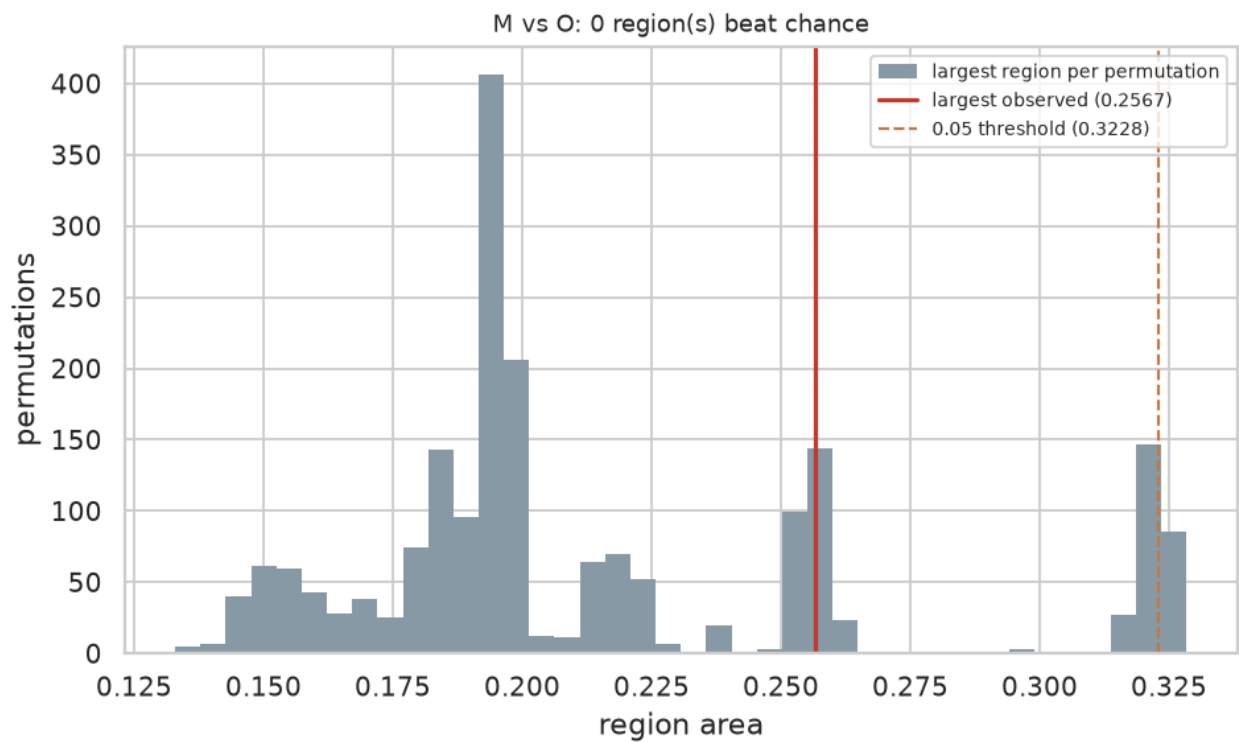


Figure 14: regions

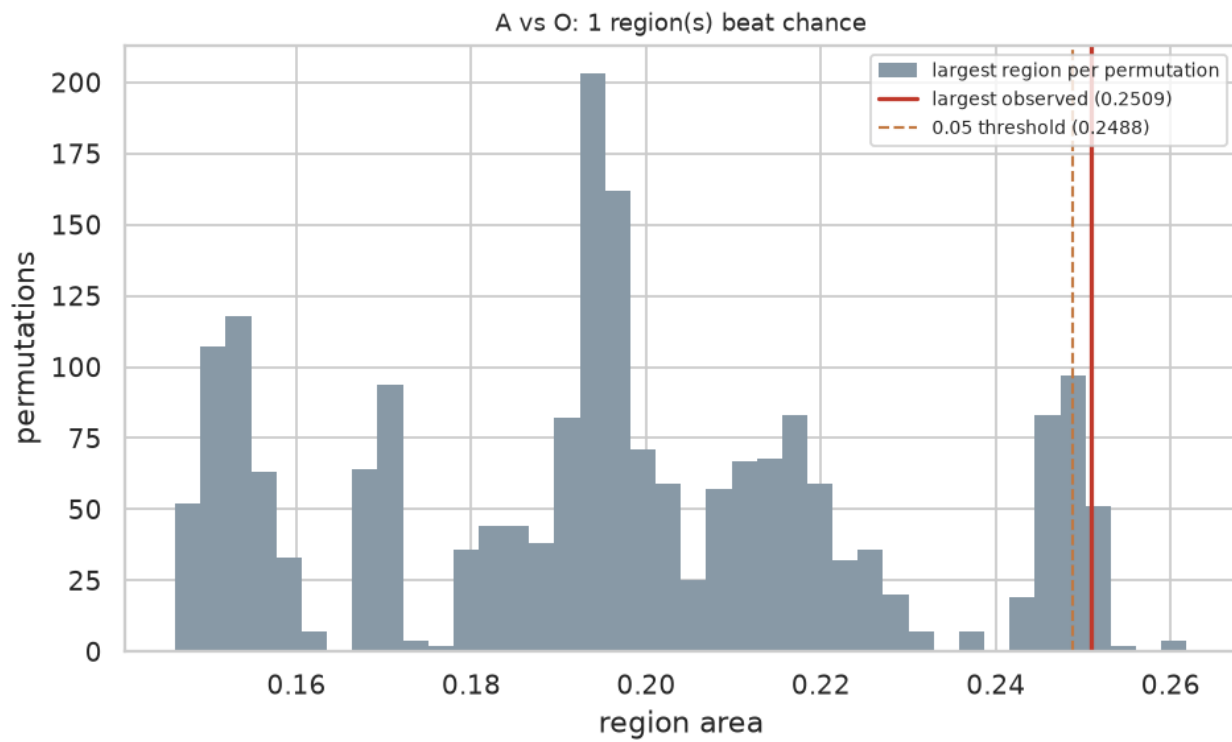


Figure 15: regions

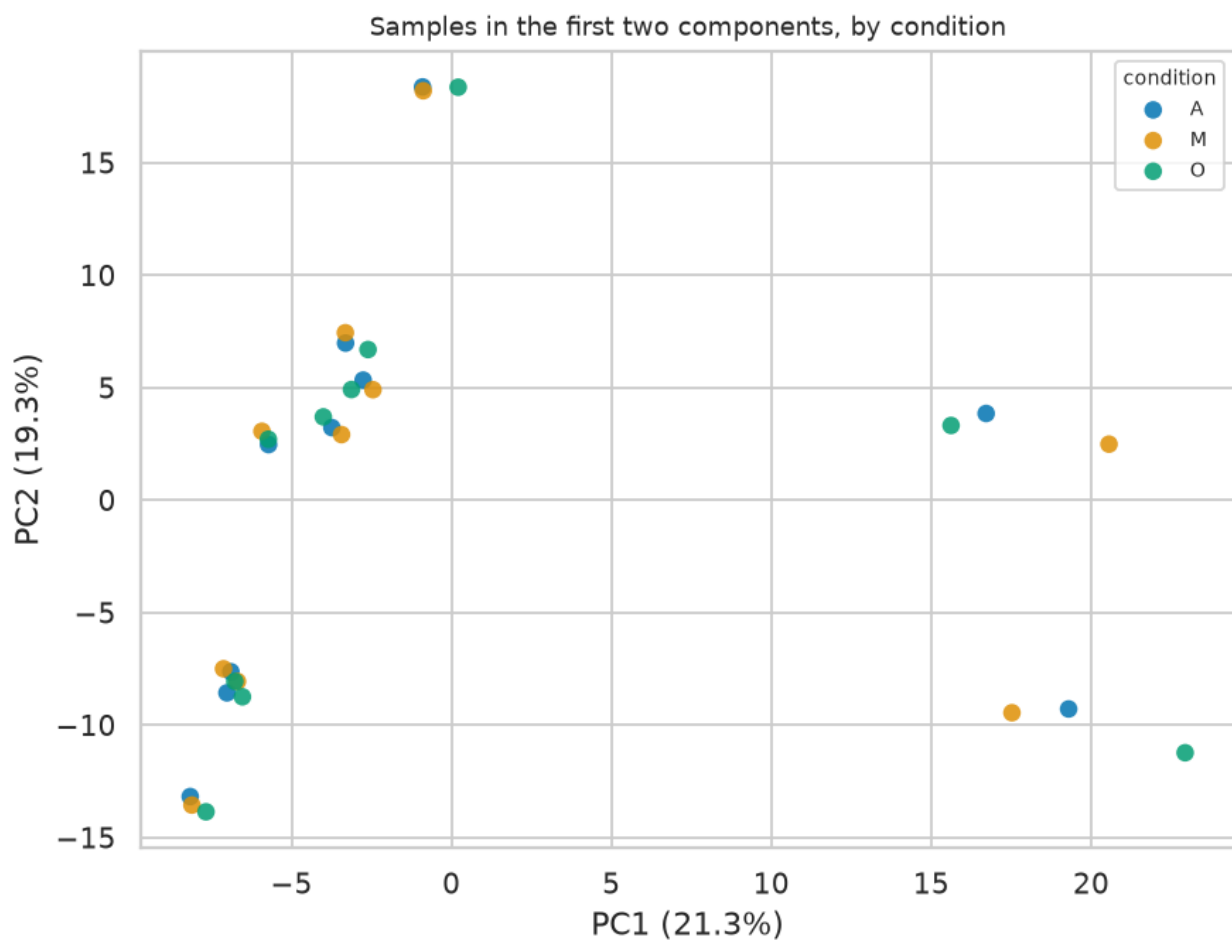


Figure 16: components

Limitations

The analysis is limited by the lack of information about the dataset, including the number of rows and columns before reshaping. Additionally, the analysis does not report any information about the class balance or the performance of the methods.

Recorded metrics

The values below are copied directly from the run record; where the narrative cites a number, this table is authoritative.

Metric	Recorded value
input rows	440110
input columns	35
r002 / rows after reshape	13203300
r002 / melted columns	30
r002 / headers parsed	13203300
r010 / limits shape a	-3.7329
r010 / limits shape b	-8.5685
r010 / bias	-0.0009
r010 / limit lower	-0.0380
r010 / limit upper	0.0362
r010 / pairs	4401100
r010 / comparison	M vs A
r010 / differences normal p	0.0000
method agreement / bias	-0.0006
method agreement / pairs	4401100
r011 / limits shape a	-3.5514
r011 / limits shape b	-8.9646
r011 / bias	-0.0015
r011 / limit lower	-0.0460
r011 / limit upper	0.0430
r011 / pairs	4401100
r011 / comparison	M vs O
r011 / differences normal p	0.0000
r012 / limits shape a	-3.6431
r012 / limits shape b	-8.7454
r012 / bias	-0.0006
r012 / limit lower	-0.0411
r012 / limit upper	0.0399
r012 / pairs	4401100
r012 / comparison	A vs O
r012 / differences normal p	0.0000
r020 / features tested	440110
r020 / subjects paired	10

r020 / significant	0
r020 / comparison	M vs A
r020 / smallest p	0.0000
r020 / detection limit	0.0600
paired feature test / features tested	440110
paired feature test / subjects paired	10
paired feature test / significant	0
r021 / features tested	440110
r021 / subjects paired	10
r021 / significant	0
r021 / comparison	M vs O
r021 / smallest p	0.0000
r021 / detection limit	0.0659
r022 / features tested	440110
r022 / subjects paired	10
r022 / significant	0
r022 / comparison	A vs O
r022 / smallest p	0.0000
r022 / detection limit	0.0626
r040 / features tested	440110
r040 / equivalent	360594
r040 / equivalent share	0.8193
r040 / margin	0.0200
r040 / median observed difference	0.0025
r040 / detection limit	0.0119
r040 / detection limit corrected	0.0600
r040 / comparison	M vs A
equivalence test / equivalent	337176
equivalence test / of	440110
r040 / of	440110
equivalence test / margin	0.0200
r041 / features tested	440110
r041 / equivalent	323248
r041 / equivalent share	0.7345
r041 / margin	0.0200
r041 / median observed difference	0.0030
r041 / detection limit	0.0131
r041 / detection limit corrected	0.0659
r041 / comparison	M vs O
r041 / of	440110
r042 / features tested	440110
r042 / equivalent	337176
r042 / equivalent share	0.7661

r042 / margin	0.0200
r042 / median observed difference	0.0028
r042 / detection limit	0.0124
r042 / detection limit corrected	0.0626
r042 / comparison	A vs O
r042 / of	440110
r050 / statistic	0.0040
r050 / p value	0.0940
r050 / permutations	2000
r050 / null median	0.0039
r050 / features	440110
r050 / subjects	10
r050 / comparison	M vs A
permutation test / statistic	0.0051
permutation test / p value	0.0155
permutation test / permutations	2000
permutation test / significant at 0.05	true
r050 / significant at 0.05	false
r051 / statistic	0.0053
r051 / p value	0.0580
r051 / permutations	2000
r051 / null median	0.0044
r051 / features	440110
r051 / subjects	10
r051 / comparison	M vs O
r051 / significant at 0.05	false
r052 / statistic	0.0051
r052 / p value	0.0155
r052 / permutations	2000
r052 / null median	0.0041
r052 / features	440110
r052 / subjects	10
r052 / comparison	A vs O
r052 / significant at 0.05	true
r060 / clusters	153239
r060 / regions found	39697
r060 / regions significant	0
r060 / largest region area	0.1121
r060 / null threshold	0.1965
r060 / permutations	2000
r060 / comparison	M vs A
differential regions / regions found	63501
differential regions / regions significant	1

r061 / clusters	153239
r061 / regions found	67169
r061 / regions significant	0
r061 / largest region area	0.2567
r061 / null threshold	0.3228
r061 / permutations	2000
r061 / comparison	M vs O
r062 / clusters	153239
r062 / regions found	63501
r062 / regions significant	1
r062 / largest region area	0.2509
r062 / null threshold	0.2488
r062 / permutations	2000
r062 / comparison	A vs O
r070 / samples	30
r070 / features used	50000
r070 / components	10
r070 / variance pc1	0.2126
r070 / associated condition	0
r070 / smallest p condition	0.0329
r070 / associated subject	9
r070 / smallest p subject	0.0007
component association / samples	30
component association / components	10
component association / factors tested	2
r070 / factors tested	2
r030 / figures	16
r030 / tables	26
r030 / measurements	147
r030 / steps reporting	23
r030 / template	academic
r030 / include plots	true
r030 / contents	16 figures and 26 tables from 23 steps

Reproducibility

Everything needed to identify and repeat this run.

Field	Value
Run	run nkQ138o18Bvr
Pipeline	Reproducing and challenging: whole blood DNA extraction meth
Started	2026-08-19 20:56:51
Finished	2026-08-19 21:10:30

Duration	818.8886 s
Compute	CPU
Figures	16

Produced on the AlkhemYst-Ai platform. Every figure and every value in the tables of this report comes from the recorded run; the full experiment lineage is available in the workspace.